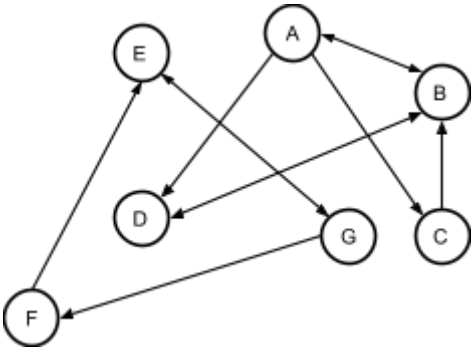
American Computer Science League

2024-2025 ● Contest 4: Short Problems Solutions ● Senior Division

1. Graph Theory

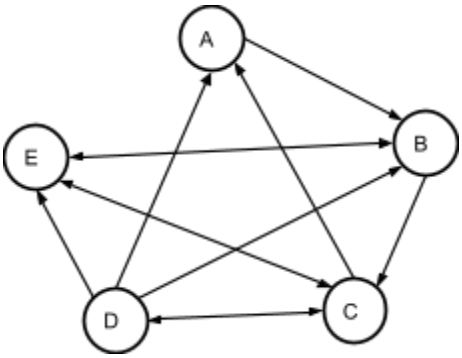
How many ordered pairs of vertices have at least 1 path of length 2 in the following directed graph? For example, (B,A) means from B to A.



A. 15
B. 17
C. 30
D. 32
E. 49

2. Graph Theory

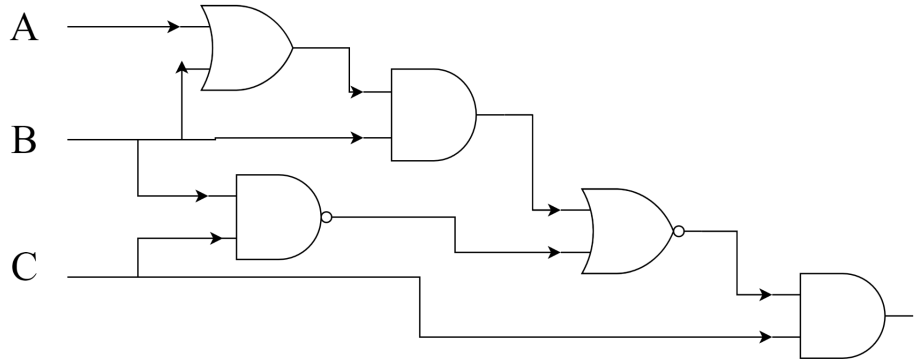
Which vertex is in the most cycles?



A. A
B. B
C. C
D. D
E. E

3. Digital Electronics

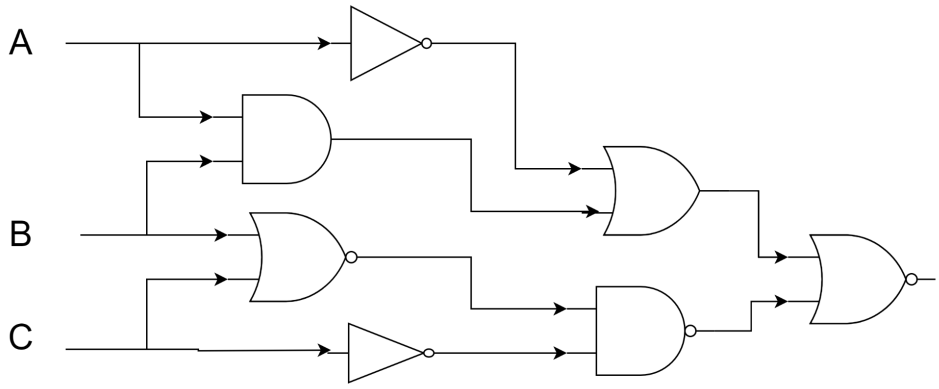
How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

4. Digital Electronics

Simplify the following expression to the fewest gates possible.



A. $\overline{A}(\overline{B + C})$
B. $A(\overline{BC})$
C. $A(\overline{B + C})$
D. $A(B + C)$
E. $\overline{A(B + C)}$

5. Assembly Language

What is the output when the following program is executed?

```
X1      DC      1
X2      DC      -2
Y1      DC      1
Y2      DC      -5
        LOAD    X2
        STORE   B
        SUB     X1
        STORE   DX
        BE      STOP
        LOAD    Y2
        SUB     Y1
        STORE   DY
        DIV     DX
        STORE   M
        MULT    X1
        SUB     Y1
        MULT    =-1
        STORE   B
STOP    PRINT   B
        END
```

A. -2
B. -1
C. 0
D. 1
E. 2

6. Assembly Language

What is the output when the following program is executed?

```
T       DC      0
NUM     DC      440
N       DC      1
TOP     LOAD    T
        SUB     NUM
        BE      STOP
        LOAD    N
        ADD     =1
        STORE   N
        MULT    N
        MULT    N
        ADD     T
        STORE   T
        BU      TOP
STOP    PRINT   N
        END
```

A. 4
B. 5
C. 6
D. 7
E. 8